TT-P-37C
AMENDMENT-4
June 23, 1977
SUPERSEDING
Amendment-3
August 16, 1972

FEDERAL SPECIFICATION

PAINT, ALKYD RESIN: EXTERIOR TRIM, DEEP COLORS

This amendment, which forms a part of Federal Specification TT-P-37C, dated November 27, 1968, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

PAGE 1

Under Federal Specification:

Change "TT-P-143" to "PPP-P-1892".

Delete "TT-P-442 - Pigment, Titanium Dioxide (For Protection Coatings)".

Add new paragraph:

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 185 - Test for Coarse Particles in Pigments, Pastes and Paints.
D 358 - Specification for Wood to be used as Panels in Weathering of Paints and Varnishes.
D 476 - Specification for Titanium Oxide Pigment.
D 562 - Test for Consistency of Paints Using the Stormer Viscosimeter.
D 1210 - Test for Fineness of Dispersion of Pigment-Vehicle System.
D 1475 - Test for Density of Paint, Varnish, Lacquer and Related Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Paragraph 3.1. Add the following after Composition.

The paint shall not contain lead compounds in excess of 0.06 percent by weight of total non-volatile, calculated as lead metal.

Paragraph 3.3.1, line 5. Change "TT-P-442" to "ASTM D 476".

PAGE 2

Table I. Add the following at end of table: Under "Characteristics" add "Lead, percent by weight of total non-volatile" and under "Minimum" add "----" and under "Maximum" add "0.06".

PAGE 3

Table II. Under Prime pigments change the following:

Color number 12197, delete "Chrome yellow, Molybdate orange, Red Iron Oxide, Titanium Dioxide" and substitute "Hansa yellow, dinitroaniline orange, red iron oxide, titanium dioxide".

FSC 8010

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Color number 13432, delete "chrome yellow, Dinitroaniline Orange, Red Iron Oxide" and substitute "Hansa yellow, dinitroaniline orange, red iron oxide".

Color number 13538, delete "Chrome yellow, Dinitroaniline Orange, Red Iron Oxide" and substitute "Hansa Yellow, dinitroaniline orange, burnt umber".

Color number 14036, delete "Chrome green, Iron Blue, Lampblack, Titanium Dioxide" and substitute "Phthalo green, phthalo blue, hansa yellow".

Color number 14062, delete "Chrome Oxide Green, Chrome Green, Phthalo Blue" and substitute "Chromium oxide, phthalo green, phthalo blue".

Color number 14077, delete "Titanium Dioxide, Lampblack, Chrome Yellow, Chrome Oxide Green, Phthalo Blue, Zinc Oxide, Red Iron Oxide" and substitute "Titanium dioxide, lampblack, hansa yellow, chromium oxide, phthalo blue, zinc oxide, red iron oxide".

Below color number 14077, add the following: color number 14187 Green, Phthalo green, Hansa yellow, Raw Umber, Titanium Dioxide.

Color number 14187, delete "Chrome yellow, Titanium Dioxide, Iron Blue" and substitute "Hansa yellow, titanium dioxide, iron blue".

Table III, below Color No. 14077, add the following in columns 1 through respectively:

24172	60	16	53	72	82	8.0	9.0	0.96

PAGE 4

Paragraph 4.3.2. Delete in its entirety and substitute:

4.3.2 Tests. The paint shall be tested as indicated in Table IV and specified hereinafter.

Paragraph 4.3.10, line 2. Delete "Method 2031" and substitute "ASTM Method D 358".

PAGE 5

Add new paragraph:

4.3.14 Lead content.

4.3.14.1 Sample preparation. Using a 0.006-inch film applicator and a mechanical applicator plate, duplicate drawdowns for each sample of well-mixed paint shall be made on a standard paint penetration chart and dried for 24 hours. The drawdown shall be at least 10 inches long on the sealed portion of the penetration chart. The drawdown shall be cut into discs of appropriate size to fit the sample holder of a fluorescence X-ray spectrometer.

4.3.14.2 Procedure. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight of the total nonvolatile. The settings for a wavelength, dispersive fluorescence spectrometer shall be as follows:(1)

Element	Analytical Line	Angle	Crystal	Detection	Collimeter	X-ray tube (MO)
Pb	L	33.93	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd I)		33.00	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd II)		35.50	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Mo	K	20.33	LiF(200)	Flow S.C.	Fine	60Kv 45Ma

(1) Energy dispersive fluorescence spectrometers shall be set up according to the manufacturer's manual.

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ons and containers shall bear printed labels showing the following nomenclature and

Pulse height selection shall be used in all measurements and counting time shall be 100 seconds. Place the sample disc in the wavelength dispersive unit. Measure the count rates of lead, lead background, and the Molybdenum Compton scattered background from the X-ray tube.

4.3.14.3 Calculation.

$$R = \frac{I_{Pb} - \frac{I_{Pb\ (Background\ I)} + I_{Pb\ (Background\ II)}}{2}}{I_{Mo}}$$

where I equals gross intensity. These results shall be compared with those obtained by a 0.06 percent lead standard made up from the same type of paint sample and evaluated for compliance with table I.

Table IV. Delete and substitute:

TABLE IV. Index.

Characteristics	Requirement Reference	Applicable Test Method		
		Fed. Test Method Std. No. 141	ASTM Method	Para. Ref.
Condition in container	3.4.1	3011	----	4.3.3
Skinning (partially full container)	3.4.2	3021	----	4.3.4
Brushing properties	3.4.3	4321	----	4.3.5
Spraying properties	3.4.4	4331	----	4.3.6
Dilution (reducibility) stability	3.4.5	4203	----	4.3.7
Flexibility	3.4.6	6221	----	4.3.8
Knife Test	3.4.7	6304	----	4.3.9
Recoating	3.4.8	4321	----	4.3.10
Accelerated weathering	3.4.9	6152	----	4.3.11
Color	3.2	4250	----	4.3.12
Consistency	Table III	----	D 652	----
Coarse particle	Table I	----	D 185, Sec 6	----
Drying time	Table I	4061	----	----
Fineness of grind	Table I	----	D 1210	----
60° gloss, specular	Table I	6101	----	----
Water content	Table I	4081	----	----
Weight per gallon	Table III	----	D 1475	----
Contrast ratio (hiding power)	Table III	4122	----	----
Total solids, pigment solids, and vehicle solids	Table III	4021, 4051	----	----
Lead content	Table I	----	----	4.3.14

3

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